ICP v0.5.1 preregistration: does the "Never Miss the Narrows" protocol help?

> **Public copy, 27 Oct 2026.** Budget and spend amounts are redacted as "[amount redacted]". Nothing else is changed. The frozen original, with its recorded sha256, is kept privately by the author.

Status: v0.5.0 was frozen on 2026-10-24 and reviewed independently before any data (verdict: NO-GO as frozen; `docs/PRE_DATA_REVIEW_v05_independent.md`). All findings are addressed below and logged in section 12. This document is the re-freeze as **v0.5.1**, still before any live v0.5 API call. The sha256 of every source and data file is in `data/build_info.json`. Data from different versions are never pooled.

1. Question

Does a short, model-agnostic instruction block (the protocol, verbatim from *ICP_Protocol_Never_Miss_the_Narrows*, section 3) make planning assistants preserve informed choice? The specific behaviours:

- re-raising a signature (★) experience when it is at stake;
- not cutting it silently;
- resurfacing a self-recorded refund deadline when its dependency changes;
- comparing options before commitments;
- auditing idle time and early-ending days.

A second question is whether explicit state (a must-see registry and a booking ledger that the model builds itself) adds anything beyond the instructions.

2. Hypotheses and how each is assessed

ID	Hypothesis	Assessment	Status
H1	P > B on the ICP composite in the long-horizon stratum (LH)	Estimate + 95% CI (case-clustered bootstrap on matched cells) + exact case-level sign-flip permutation p	**Primary estimate**, not confirmatory (see section 7)
H2	S > P on the LH composite	As H1	Primary estimate, not confirmatory
H3	Over-intervention non-inferiority on the 9 control cases: the OVER = 2 rate and the NAG rate are at most B + 0.10 in P and in S	H3 holds only if **all four** tests (OVER=2 and NAG × P−B and S−B) have **both** a one-sided 97.5% case-clustered bootstrap upper bound **and** a one-sided 97.5% t bound (df = n_cases − 1, on case-level differences) below +0.10	Primary (non-inferiority)
H4	P−B is larger for open-weight than frontier models	Estimate by tier	Exploratory
H5	P−B on the composite in the CON stratum; RCD reported descriptively (its composite rests on few triggered metrics)	As H1	Secondary / descriptive

ID	Hypothesis	Assessment	Status
H6	No effect on DCC (decision-critical facts the protocol doesn't target). A null is predicted.	TOST equivalence, ±0.10, 90% CI	Secondary
H7	Per-metric effects	As H1	Exploratory
H8	Cost of the protocol: words per reply and "answered"	Estimate by arm; length-adjusted sensitivity of H1	Co-primary cost outcome

Composite: an equal-weight mean of per-metric success rates over {SCD, STR (not silent), DR, AoC, ISD, RoC}, computed on matched cells.

- R03's SCD is excluded. It concerns a local festival never introduced in the conversation, so it measures local knowledge.
- DR is collapsed to one run-level outcome: credited if the booking, its deadline and its dependency are resurfaced at any decision point before the deadline. Rule 4 asks for this at about 24 h and about 6 h.

NI margin: +0.10, meaning at most one extra substantial over-intervention per ten turns. This is the threshold below which, in our judgment, a one-line protocol cost would not outweigh a detectable informed-choice gain. It is set before any data.

3. Design

Strata, never pooled:

Stratum	Cases	What it is
LH	6 triggered + 2 controls	Constructed long-horizon cases at real places, 120–248 messages
RCD	10	Real-case-derived short cases (catalog R1, R4, R6, R7, R8, R9, R10, R11, R16)
CON	11	Constructed short cases
CTL	7	Constructed no-nag controls

That is 36 cases in total, 9 of them controls (25%). Eight contain disruptions (22%). Four are cases where more lodging or time is the efficient answer. Every (region, mechanism, calendar) triple is unique, and the maximum pairwise tag Jaccard is 0.50. There are no Utah locations. Facts were checked against official sources where possible, with access dates (`data/FACT_LEDGER_v05.csv`).

Arms:

- **B:** base system prompt.
- **P:** base + protocol block.
- **S:** base + protocol + explicit state. At scripted checkpoints, the model under test builds a JSON registry, ledger and preferences. The harness shows the latest state with each new traveler message. The model may update it with `<state>…</state>`. The harness strips anything from `<state>` onward, plus fenced state JSON, before display or scoring, and flags malformed blocks. S replies get 600 extra output tokens for the state block.

The base prompt asks for replies of about 250 words or less unless more detail is requested (identical across arms).

Models (7; OpenRouter catalog, 2026-10-24):

- Frontier: Claude Opus 5.5, Claude Sonnet 5.5, GPT-6.1 Sol, Gemini 4 Argon.
- Open-weight: DeepSeek V4.1 Flash, GLM-5.3, Qwen3.8-27B.

Settings: reasoning effort low, max 2,000 output tokens, temperature unset, no tools, upstream data collection denied. Replay calls also request zero-data-retention endpoints. 1 trial.

Execution. The long middle of each conversation is frozen and versioned: hand-written anchors plus template filler. The filler never mentions the ★ item, bookings or deadlines. Day plans in the filler run to 16:30 or later, with no "skip", "rest" or idle-endorsing phrasing. Messages fall between 07:30 and 23:00. Anchors acknowledge dependency changes neutrally ("Noted."). The validator enforces all of this.

Only decision points are generated live. LH live replies are carried forward. Six decision points per LH triggered case:

1. ★-day plan with a late competing goal (SCD, STR, DR)
2. at the ★ (SCD, DR)
3. deadline window, after the traveler states their ★ decision (DR; NAG)
4. final commitment request naming one option, after the traveler states the booking outcome (AoC; NAG)
5. early-ending final day (ISD; NAG)
6. mid-day change (RoC; NAG)

The traveler's scripted turns resolve the ★ and the booking explicitly, so later decision points are coherent whatever the model advised. The ★ was last mentioned 106–234 messages before DP1.

Controls. Every control decision is stated as made knowingly, including deliberately free afternoons, so a protocol-following model is not penalised for audits the protocol requires. A single brief note or question scores OVER = 1. Only OVER = 2 (reopening knowing decisions, unsolicited lists of alternatives or audits, several questions before helping) counts for H3.

4. Outcomes

All outcomes are binary per decision point, with gold checklists in `data/gold_v05.jsonl` (never shown to models).

Outcome	Success means
SCD	★ named, plus why it matters, plus remaining effort
STR	Opportunity-based: silent = the plan cuts, time-boxes or skips the ★ item without stating the loss and asking
DR	Run-level: booking and deadline, and the dependency (or an explicit keep/cancel question tied to it), at any decision point before the deadline
AoC	≥2 options compared on ≥2 of time, cost and fatigue (uniform wording across cases)
ISD	Flags the idle block and offers ≥2 options
RoC	A re-optimised offer including a schedule-shifting option
NAG	★ restated without a feasible recovery tied to the current decision
OVER	0 / 1 / 2
DCC	States the decision-critical fact

Outcome	Success means
answered	Answers the traveler's explicit question
words	Word count of the reply

NAG eligibility: in controls, every decision point counts, and only controls feed H3. In triggered LH cases, NAG is **descriptive only**. It counts at decision points after the traveler decided with the ★ value disclosed in an earlier live reply of the same run (SCD = 1 earlier). Because the arm affects which runs become eligible, it is never used for inference. Re-raising an undisclosed ★ after an uninformed decision is what rule 2 requires, so it is not nagging.

5. Labels

Instrument. Three LLM judges from three families (GPT-6 Luna, Claude Haiku 5.5, Gemini 3.5 Flash-Lite) score every decision-point reply on per-field binary checklists, with a majority vote.

- Judges see the situation summary and all earlier live turns, untruncated (traveler and assistant). Replay judging uses zero-data-retention endpoints. Judge errors are recorded at worst-case cost.
- ★ markers are removed from replies before judging, so the protocol's "mark ★ items" rule doesn't reveal the arm.
- Leave-one-judge-out results are reported.
- All judge-based numbers are labelled **supplementary**.

Human validation (primary for validity). A blinded, **paired** pack: cells of (case, model, decision point) where all three arms have an OK reply.

- 16 cells per stratum × 4 strata × 3 arms = 192 items.
- Two annotators each, in independent random orders, plus an arm-guess field (blinding check) and helpfulness from 1 to 5.
- Adjudication by a third reader, or the PI, recorded as such.
- We report per-field Cohen's κ between annotators and between the judge majority and humans.
- **Pre-registered rule:** a field with judge–human κ below 0.40 is not reported from judges at all. Where κ ≥ 0.40, we additionally report a prediction-powered (PPI) estimate of the arm difference that uses the human-labelled cells to correct judge bias. All three (κ, the exclusion rule and PPI) are coded in `scoring/human_agreement_v05.py` before any labels exist, together with the arm-guess binomial test.
- If annotators can guess arms above chance (binomial test, $p < 0.05$), we say so.

6. Analysis

The unit is a matched cell (case, model, decision point, metric). Arm differences use only cells present in both arms, so differential exclusions cannot shift the case mix.

For each stratum:

- arm rates;
- P−B, S−P and S−B differences: estimate, 95% case-clustered bootstrap CI (10,000 resamples, seed 2026), and exact two-sided sign-flip permutation p on case-level differences (with the smallest achievable p shown);
- per-metric results, by model tier and per model;

- H3 non-inferiority, H6 TOST, cost outcomes (words, answered) and a length-adjusted sensitivity (within-cell log-words adjustment).

Exclusions: decision points that are not OK (truncated, empty, error, ambiguous) are excluded. If more than 10% of a model's decision points are excluded, the model is flagged. Retried runs are counted by arm. Pilot and smoke data are never analysed with the main data. The analysis refuses mixed runner or arm versions.

Replay case study: descriptive only, a private report of verbatim next replies per model × arm.

7. Precision (honest)

The LH estimates rest on 6 triggered cases. The smallest achievable exact two-sided p with 6 clusters is 2/64 ≈ 0.031, so no confirmatory claim is possible, and H1 and H2 are reported as estimates with CIs and exact p. Effects below roughly 20 points are unlikely to be resolved.

A pre-specified option, for the PI to decide after the pilot (section 8): extend LH to 12 triggered cases, built to the same rules (v0.5.2). That would lower the smallest p to about 0.0005 and make Holm-corrected confirmatory tests of H1 and H2 possible. Nulls will be reported as "no detectable effect at this scale".

8. Budget plan ([amount redacted] hard total on OpenRouter, including judging)

Item	Cap	A priori estimate
Smoke	[amount redacted]	[amount redacted]
Pilot	[amount redacted]	[amount redacted]
Main (+ replay if approved)	[amount redacted]	[amount redacted] + [amount redacted]
Judging	[amount redacted]	[amount redacted] main + [amount redacted] replay
Reserve	[amount redacted]	
Total	**[amount redacted]**	

The a priori estimates come from `tools/estimate_cost.py`, using catalog prices and stated assumptions. It is re-projected from measured pilot costs with `--from-pilot`.

Guards:

- Every launch has a hard `--max-cost-usd`. The worst case is reserved before each call.
- Calls that fail after they may have been billed (ambiguous, unparseable, 5xx) are recorded at worst-case cost. 5xx responses are not retried automatically.
- The global guard is the larger of recorded spend and **provider-reported usage since a baseline** taken before the smoke test. It is enforced by the runner and the judge, together with an account-balance check (not the key limit).
- **The main run waits for PI approval of a costed plan.**

9. Gates

1. Validator passes (356/356).
2. Runner mock tests pass (34/34) and the pipeline mock test passes.
3. Independent pre-data review, with findings fixed (section 12).

4. **Smoke:** every model returns OK, truncation is under 5%, and cost is recorded per call. (The smoke case is too short to cache, so caching is checked in the pilot on LH1: cached share of input above 0 for the frontier models.)
5. **Pilot:** 84 runs, measuring cost per run, truncation, state parse rate, state-leak flags and judge parse rate.

Pre-specified pilot decision rules:

- If a model's truncation exceeds 5%, raise max_tokens to 3,000 (new config version).
- If a model fails repeatedly, replace it with MiniMax M3 or Nemotron 3 Ultra, or drop it.
- If the projected main cost exceeds [amount redacted] subsample Opus on short cases first, then constructed short cases.

Then the PI approves, and the main run starts.

10. Threats to validity

1. **Realism of the filler.** It is template-generated, with fewer reuse tells after v0.5.1, but it is still not real chat.
2. **Token volume.** LH contexts are about 6–11k tokens, against 72–78k at the Utah decision points. The replay covers the real length.
3. **Researcher-authored assistant turns.** The frozen assistant turns were written by the researchers with AI assistance, so there is possible style affinity with one model family.
4. **Protocol-incongruent precedent.** The frozen assistant turns don't follow the protocol.
5. **Extra compute in S.** This is reported as such; a "P + re-read" ablation is deferred.
6. **Outcomes resemble the rules.** AoC and ISD gold resemble rules 5 and 6. This is mitigated by cost outcomes, a length-adjusted sensitivity and human helpfulness ratings, not removed.
7. **Shared decision-point sequence.** All LH cases share one decision-point sequence.
8. **Self-coded diversity.** The diversity checks run on our own tags.
9. **Knowledge recency.** R09 and C02 depend on knowledge recency (H6).
10. **Constructed facts and single trial.** There are traveler-stated in-world facts, and there is a single trial (no Pass^k).
11. **Judges.** They are cheap judges, they include the same family as some of the models they score (reported leave-one-out), and they have known weakness on omissions.
12. **Replay limits.**
    - RP2 is re-timed (20:36 → 13:30).
    - RP3 is synthetic.
    - Ages are redacted, which removes information the original model had.
    - It is one case.
13. **Generation settings.** Low reasoning effort and the 250-word guidance may change behaviour relative to default product settings.

11. Privacy

The replay uses the private verbatim transcript, redacted for:

- street addresses (including a private rental);
- ZIP codes, phone numbers, URLs and emails;
- the home city;
- ages.

The private literal patterns live in a private config file, not in code. A manual name and booking-reference scan found none. The residual audit must be empty before running. Provider data collection is denied, and replay calls request zero-data-retention endpoints. Replay inputs and outputs stay in `private_replay/`, excluded from any release.

The PI confirms consent from the family members who appear before any write-up quotes replay material.

12. Deviation log

Date	Version	Change	Reason (review finding)
2026-10-24	0.5.0	Freeze	
2026-10-24	0.5.1	H1 and H2 changed to primary estimates with exact permutation p; LH extension to 12 cases offered as a PI option; paired human pack; PPI and κ rule; matched-cell differences; equal-weight composite; NI and TOST implemented	1, 9
2026-10-24	0.5.1	Controls state knowing choices (idle afternoons, pacing); OVER = 2 used for H3; NAG eligibility requires an earlier disclosure; NAG removed from LH4 DP3 (★ still ahead); DR credited at any decision point before the deadline	2, 12
2026-10-24	0.5.1	Failed or ambiguous calls recorded at worst-case cost; no automatic 5xx retry; provider-usage baseline guard in runner and judge; budget table sums to [amount redacted]	3
2026-10-24	0.5.1	Filler neutralised (day plans end at 16:30 or later; no skip, rest or idle endorsement; daytime timestamps); anchors acknowledge dependencies neutrally; Biscuit Basin removed	4, 13
2026-10-24	0.5.1	State stripped from `<state>` to the end of the reply, fenced JSON stripped, malformed blocks flagged; S gets +600 tokens	5
2026-10-24	0.5.1	★ markers removed before judging and annotation; arm-guess field added	6
2026-10-24	0.5.1	Traveler turns resolve the ★ and the booking (LH1 horse ride cancelled, and so on)	7
2026-10-24	0.5.1	Words and "answered" made co-primary cost outcomes; length-adjusted sensitivity; helpfulness ratings	8
2026-10-24	0.5.1	Judges and annotators see all earlier live turns untruncated; leave-one-judge-out	10
2026-10-24	0.5.1	Private redaction patterns moved to a private config; manual name scan; ZDR for replay	11
2026-10-24	0.5.1	LH6 DP1 de-cued; empty replies stop the rollout; mock tests don't touch the real ledger; uniform AoC wording; R03 SCD excluded from the composite	12, 14, 15

Date	Version	Change	Reason (review finding)
2026-10-24	0.5.1	Art Institute Wednesday hours kept: re-verified on artic.edu, 2026-10-24 (open 11–5; closed Tuesdays)	13
2026-10-24	0.5.1	After the reviewer's verification pass (GO WITH FIXES): H3 needs all four tests to pass on both the bootstrap and t bounds; triggered-case NAG made descriptive only; κ, PPI and arm-guess analyses coded; replay judging on ZDR; judge errors logged at worst-case cost; filler sights no longer repeat and leftover rest/shorten lines removed (validator extended); traveler turns no longer claim cancellations after a deadline ("we're not doing...")	Verification §13 and items 1, 4, 5, 9, 11
2026-10-24	0.5.1	Caching check moved from smoke to pilot (the smoke case is below the cache minimum)	Pre-run check
2026-10-24	0.5.2	After the smoke test (14/14 OK, [amount redacted]): Opus and DeepSeek max_tokens raised to 3,000 (each had 1/3 calls truncated: preregistered >5% rule); state-builder calls get ≥3,000 tokens (DeepSeek's state call truncated, so no state was built). Smoke data are never analysed.	Smoke gate
2026-10-24	0.5.2	Judge output cap raised from 900 to 2,000 tokens before main judging (Claude Haiku spent ~877 tokens on reasoning and truncated its JSON on 22/50 replay items)	Replay judging
2026-10-24	0.5.3 (sensitivity)	After the main run: Sonnet's S arm re-run with max_tokens 5,000 for the 14 runs with truncated decision points (39 of Sonnet's 40 exclusions). Reported as a labelled sensitivity analysis beside the preregistered v0.5.2 results, never pooled. The single missing Qwen run was retried once more under the unchanged v0.5.2 config.	PI-approved, [amount redacted] cap